Dynamics of Electrical Machines and Drives

Non-linearity problems and Anti-windup

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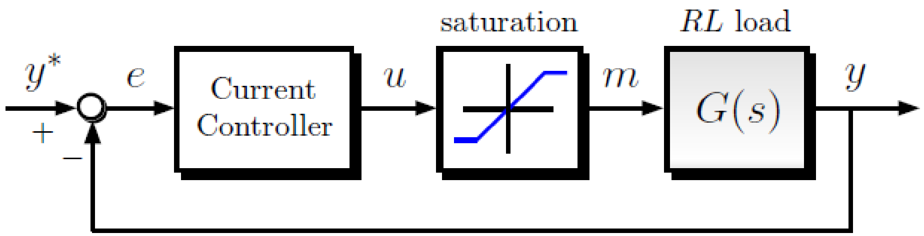
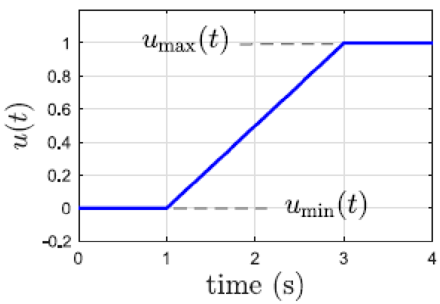
Non-linear effects
Actuator saturation



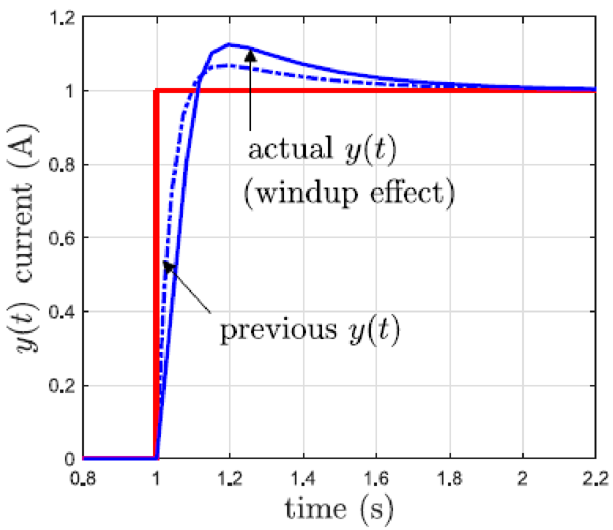


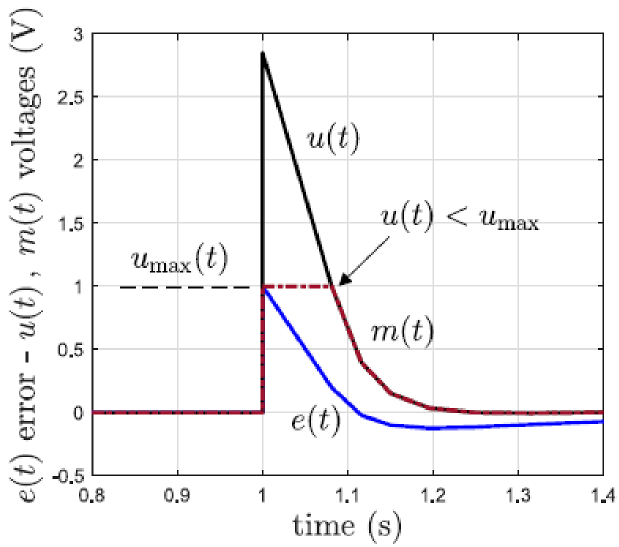
- A linear controller is simple to design, but the performances are good as long as the dynamics remain *close* to linear theory
- Nonlinear effects require care, such as **actuator saturation**
  - Saturation can be defined as the static nonlinearity

$$\text{sat}\,(u(t)) \begin{cases} u_{\min}(t) & \text{if} \quad u(t) < u_{\min}(t) \\ u(t) & \text{if} \quad u_{\min}(t) < u(t) < u_{\max}(t) \\ u_{\max}(t) & \text{if} \quad u(t) > u_{\max}(t) \end{cases}$$

$u_{\min}(t)$ and $u_{\max}(t)$ are min and max allowed actuation signals (e.g. $\pm 1\,V$ for a voltage supply)

The Windup problem
Integrating error

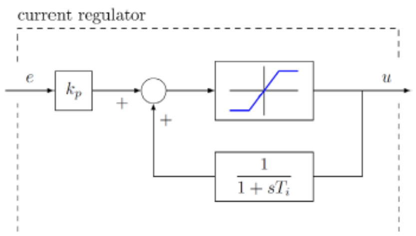


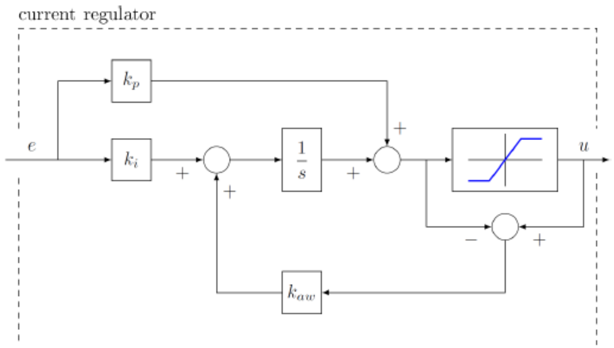


- The output $y(t)$ takes a longer time to achieve steady-state due to "*windup*" of the integrator (higher peak response)
- If $e(t)$ increase, $u(t)$ increase even if $m(t)$ is saturated to u_{max}, the PI **keeps integrating the tracking error** $e(s)$ → producing $u(t) \neq m(t)$
- If $e(t)$ change sign we have to wait until $u(t) < u_{max}$ before to come back into linear region $m(t) = u(t)$, this means wait for *integral discharge*

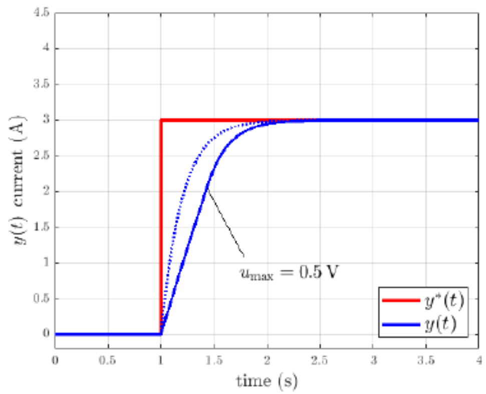
Anti-windup Techniques
Saturation model

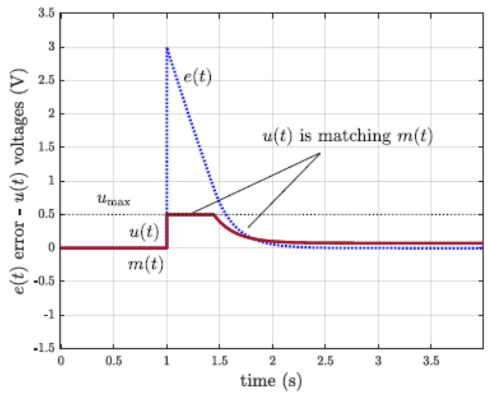
The solution to the windup issue is the inclusion of saturation information into the controller structure.





Back-calculation antiwindup schemes





No effect when the actuator is not saturating. The windup problem is avoided thanks to the back information

Exercises

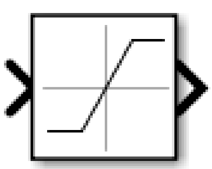
Saturation model

Let's still consider the same case of the previous lesson.

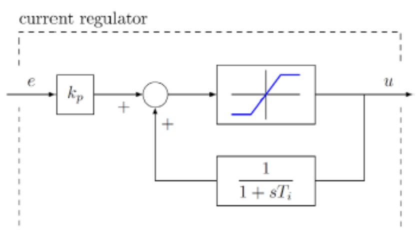
Let's assume an L=0.1H and R=0.025

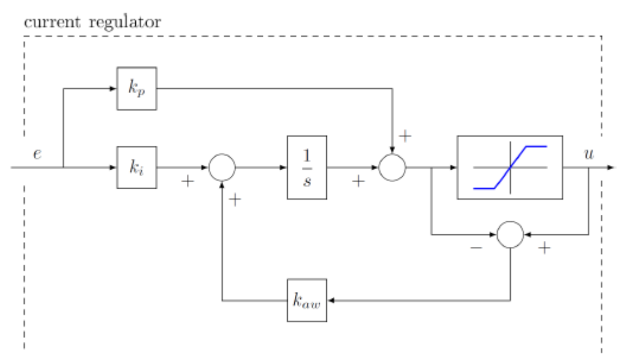
The transient has a time constant τ= L/R and this means that the constant value will be reached at T = 5τ.

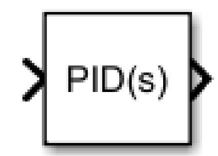
On Simulink, insert at first the saturation limit of the PI controller and then try to implement the anti-windup schemes to see what happens.



Saturation







PID Controller

Exercises
Final model model with simscape

To exploit more the Simulink possibilities, let's try to solve the same exercise using the simscape library.